

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460334.8777 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.056
Transit depth (Rp/Rs)^2	2.4 +/- 1.75 [%]
Orbital Inclination (inc)	86.75 +/- 2.86
Airmass coefficient 1 (a1)	1.797 +/- 0.09
Airmass coefficient 2 (a2)	-0.43 +/- 0.13
Scatter in the residuals of the lightcurve fit is	4.72 %
Best Comparison Star	#2 - [405, 61]
Optimal Aperture	2.18
Optimal Annulus	8.54
Transit Duration (day)	0.0587 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.056 vs 0.1594 (deviation 0.08σ)
Duration fit vs published	0.0587 d vs 0.0512 d (deviation 1.41σ)
Mid-time O–C	6.1 min
Depth SNR	2.8
Residual scatter	4.72 %

Light curve

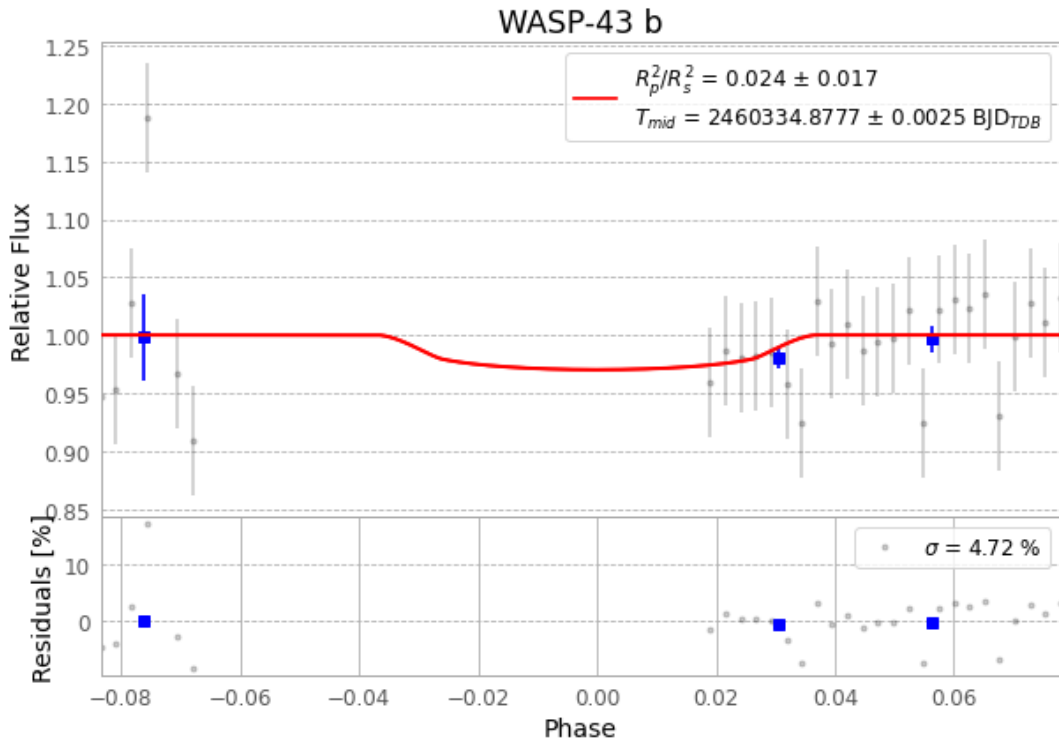


Figure 1 — FinalLightCurve_WASP-43 b_2024-01-25.png (SHA-256 313c369e1f10d93a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [483, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ca7d829681393569...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), WASP-43240125071815.FITS → WASP-43240125102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240125.log (18c54f515148...), FinalLightCurve_WASP-43 b_2024-01-25.png (313c369e1f10...), FinalLightCurve_WASP-43 b_2024-01-25.csv (3aef38e30ddb...)

Compute resources

Wall time 265.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 16:40Z.